



Antonello Giorgio

AI Engineer specialised in Audio & Music

Barcelona, Spain

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LinkedIn Profile

Personal Portfolio

GitHub Repo

TECHNICAL EXPERTISE

Machine Learning & AI

Traditional and modern machine learning methods. Experience designing and training complex neural architectures, including Transformers and Diffusion models. Skilled in experiment tracking, logging, and model fine-tuning.

Audio & Music Technology

Extracting meaningful features from audio through digital signal processing. Developing models capable of generating or modifying sound and analyzing symbolic music. Adapting speech recognition systems to handle singing.

Software Engineering

Programming in Python, C++, Java, and C. Comfortable with collaborative development workflows using Git, Docker, and Linux environments.

LANGUAGES

Italian — Native

English — Advanced (C1)

Spanish — Intermediate (B2)

SOFT SKILLS

- ✓ Systems thinking & abstraction
- ✓ Cross-disciplinary synthesis
- ✓ Clear written & oral communication
- ✓ Collaborative team-working

EDUCATION

MSc in Sound and Music Computing

Universitat Pompeu Fabra (UPF) · Barcelona, Spain

Sep 2025 – Aug 2026

Organised by the Music Technology Group, a leading audio research centre in Europe. The curriculum provided a deep dive into the intersection of computer science and acoustics. I gained hands-on experience building deep learning systems for audio generation and analysis, established a strong foundation in digital signal processing, and explored computational methods to extract musical features from raw audio. The project-based approach allowed me to connect theoretical machine learning concepts with real-world audio challenges. My thesis focuses on automatic lyric transcription and alignment for expressive singing (See ExpressiveALT project below).

MSc in Artificial Intelligence and Robotics

Sapienza University of Rome · Italy

Jan 2022 – Jul 2025

Advanced coursework covering classical Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Robotics. The programme placed a strong emphasis on project-based learning: most courses concluded with a group research project and a formal presentation to faculty, building both technical depth and the ability to communicate complex ideas clearly. My final thesis was developed during a research stay at Imperial College London (see Experience below).

BSc in Computer and Control Engineering

Sapienza University of Rome · Italy

Sep 2018 – Dec 2021

Broad engineering foundations covering algorithms, software engineering, control theory, electronics, and applied mathematics. Practical programming experience across Python, C, Java, C++, and various web technologies, with early exposure to collaborative development workflows using version control.

EXPERIENCE

Graduate Student Researcher

Imperial College London · Dept. of Electrical & Electronic Engineering

Sep 2024 – Mar 2025

Visiting researcher under the supervision of Prof. Danilo Mandic to develop my master's thesis. I designed a neural network that processes brainwave recordings (EEG) to understand a listener's emotional response to music. The system combined a Transformer encoder with a Graph Convolutional Network, and used a fine-tuned Large Language Model to interpret the complex brain signals. When evaluated on a standard benchmark dataset, the model achieved nearly **67% accuracy in predicting emotional arousal** and **65% in predicting emotional valence**, surpassing previous benchmark results in the field.

SELECTED PROJECTS

Full list at agiorgio99.github.io

ExpressiveALT — Automatic Lyrics Transcription for Expressive Singing

Standard automatic speech recognition models often fail when applied to singing due to extreme pitch and rhythm variations. To address this, I designed a pipeline that evaluates existing transcription models and synthetically generates training data by simulating specific vocal techniques, such as vibrato and falsetto. I then fine-tuned the transcription model on this expanded dataset. A controlled study demonstrated that the synthesized data was a highly effective substitute for real human recordings, helping the model reduce its **word error rate from 15.8% to 11.3%** on unseen test data.

Monteverdi's Madrigals — Computational Stylistic Analysis

Working in a small group, we used Python to analyse the complete scores of Monteverdi's nine books of madrigals to test a long-standing claim: that his style shifted dramatically around Book 5, moving toward a freer, more emotionally expressive approach. We designed three measurable features to capture this shift, grounding each one in music theory literature to ensure they were meaningful rather than arbitrary. The results **confirmed the break at Book 5**, backing expert opinion with quantitative data for the first time.

Tennis Stroke Quality Assessment

I trained a neural network to score how well a tennis player executes a stroke by comparing their movement to professional-level references across thousands of videos. The main challenge was that strokes vary enormously in timing and style; the model had to learn a general notion of quality rather than memorising specific examples. The final system provided scores that were on average **11% closer** to expert evaluations than the previous best method on the same dataset.